# SIMATS ENGINEERING

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**Machine Learning**

**Experiment -13**

## Implement Iris Flower Classification using KNN

## Code:

```
# Car Price Prediction using Linear Regression

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Load dataset
df = pd.read_csv("play_tennis (1).csv")

# Display dataset
print("Dataset:")
print(df.head())

# Handle categorical columns
encoder = LabelEncoder()

for column in df.select_dtypes(include=["object"]).columns:
    # Only encode non-target categorical features
    if column != 'play':
        df[column] = encoder.fit_transform(df[column])

# Separate features and target
X = df.drop("play", axis=1) # Changed from "Price" to "play"
y = df["play"] # Changed from "Price" to "play"

# Convert categorical target to numerical for Linear Regression
y = y.map({'Yes': 1, 'No': 0})

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Predict 'play' values
y_pred = model.predict(X_test)

# Evaluate model
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("\nLinear Regression Model for Play Prediction") # Updated title
```

```
print("-----------------------------------------")
print("Mean Absolute Error:", mae)
print("Mean Squared Error:", mse)
print("R2 Score:", r2)

# Display actual vs predicted 'play' values
result = pd.DataFrame({
    "Actual Play": y_test.values, # Updated column name
    "Predicted Play": y_pred # Updated column name
})

print("\nActual vs Predicted Play:") # Updated title
print(result)
```

OUTPUT:

```
Dataset:
  day   outlook  temp humidity    wind play
0  D1     Sunny   Hot     High    Weak   No
1  D2     Sunny   Hot     High  Strong   No
2  D3  Overcast   Hot     High    Weak  Yes
3  D4      Rain  Mild     High    Weak  Yes
4  D5      Rain  Cool   Normal    Weak  Yes

Linear Regression Model for Play Prediction
-----------------------------------------
Mean Absolute Error: 0.5504262000897261
Mean Squared Error: 0.3189280775733607
R2 Score: -0.43517634908012304

Actual vs Predicted Play:
   Actual Play  Predicted Play
0            1        1.389906
1            1        0.437236
2            0        0.698609
```